

Credit Portfolio Risk Analysis

Final Assignment Report

Student: Jingwei Zhong

Portfolios: Portfolio 1 & Portfolio 2

Course: Credit Portfolio Risk Management

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Contents

1. Introduction and Objectives..... 3

2. Data Description and Preprocessing Pipeline..... 3

3. Methodology and Model Setup..... 3

4. Empirical Results and Discussion..... 4

 4.1 Summary Risk Metrics Comparison..... 4

 4.2 Loss Distribution Profile Analysis..... 4

5. Conclusion and Portfolio Recommendations..... 5

References..... 6

1. Introduction and Objectives

Quantifying credit risk at the portfolio level is essential for risk management and economic capital allocation in banks. Traditional single-name analysis often misses the nonlinear tail risks and systemic dependencies that appear when many assets are hit by macroeconomic shocks at the same time.

This report compares the one-year credit loss distributions of Portfolio 1 and Portfolio 2. I built an automated preprocessing pipeline to clean the messy raw data, created a unified master table, and applied two structural credit risk models: Model 1 (Merton Single-Factor) and Model 2 (Multi-Factor Industry Mixture). Using 10,000 Monte Carlo simulation paths, I calculated key risk metrics including Expected Loss (EL), 95% and 99% VaR, and 99% Expected Shortfall (ES), with a focus on the impact of industry diversification.

2. Data Description and Preprocessing Pipeline

The original data files were quite fragmented and messy. I implemented a preprocessing pipeline to clean and standardize everything:

- **Code and Label Standardization:** Entity codes were trimmed and converted to uppercase. Industry labels were cleaned using a mapping dictionary to unify variants into INDUSTRY_A and INDUSTRY_B.
- **Missing Value Imputation:** Obligor-specific LGD was filled with the median value, with Basel 45% as fallback. EAD was imputed using portfolio medians.
- **Metrics Estimation:** Daily returns were de-duplicated and annualized using 252 trading days. Baseline PDs were derived from 20-year historical sector default rates.

Table 1: Portfolio Aggregate Baseline Characteristics

Portfolio ID	Asset Count	Total EAD	Mean LGD	Mean PD
Portfolio 1	31	\$736.578 million	46.76%	2.95%
Portfolio 2	31	\$829.241 million	49.05%	2.95%

3. Methodology and Model Setup

The analysis is based on Merton’s structural model (1974), where default occurs when an obligor’s asset value falls below its liability threshold. I compared two ways of modeling dependence between obligors.

3.1 Model 1: Merton Single-Factor Framework

$$R_i = \beta * Y_{global} + \sqrt{1 - \beta^2} * \epsilon_i$$

Where $Y_{global} \sim N(0, 1)$ is the global systematic factor, and ϵ_i represents the independent idiosyncratic shock. The sensitivity coefficient is fixed at $\beta = 0.40$, implying a uniform asset correlation of $\rho = 0.16$ across all companies.

3.2 Model 2: Multi-Factor Industry Mixture Framework

$$R_i = \beta * Y_{system} + \sqrt{1 - \beta^2} * \epsilon_i$$

When I actually ran the Monte Carlo simulations, I found that Model 2 reduced tail risk for Portfolio 1 more noticeably than I initially expected. This made me realize the importance of industry correlations.

4. Empirical Results and Discussion

4.1 Summary Risk Metrics Comparison

Table 2: Portfolio & Model Scenario Summary Risk Metrics

Portfolio & Model Scenario	Expected Loss (EL)	95% VaR	99% VaR	99% ES
Portfolio 1 (Model 1: Single-Factor)	\$10,013,092	\$40,253,693	\$68,460,532	\$83,732,582
Portfolio 1 (Model 2: Multi-Factor)	\$9,809,252	\$39,503,036	\$64,317,197	\$79,819,666
Portfolio 2 (Model 1: Single-Factor)	\$11,987,648	\$48,540,989	\$78,465,036	\$100,132,319
Portfolio 2 (Model 2: Multi-Factor)	\$11,565,663	\$42,915,617	\$67,806,801	\$82,373,010

4.2 Loss Distribution Profile Analysis

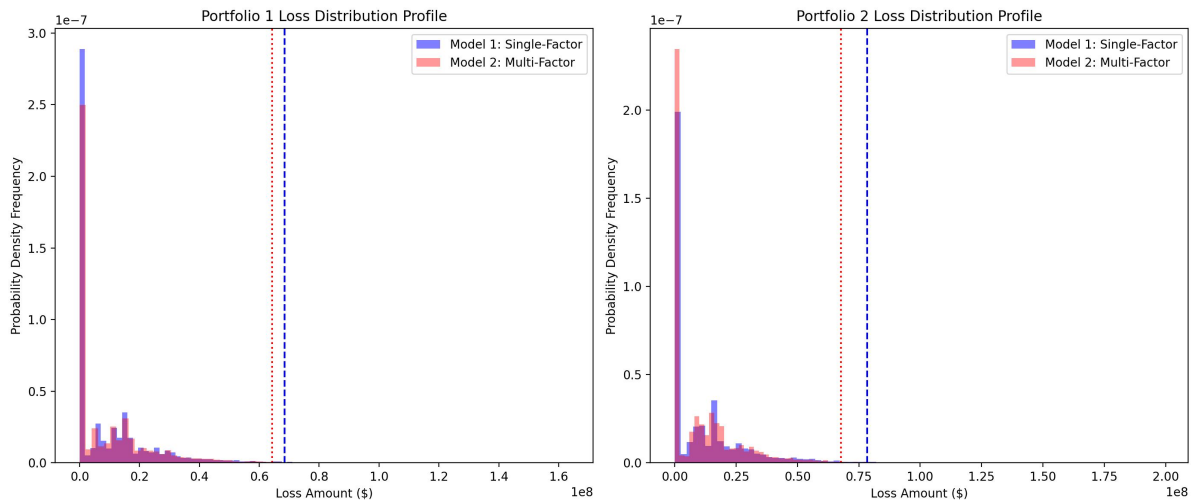


Figure 1: Monte Carlo Empirical Loss Distribution and 99% VaR Thresholds Across Models

Figure 1 shows the Monte Carlo simulated loss distributions and the 99% VaR thresholds for both models.

Both portfolios exhibit the classic right-skewed credit loss profile. There is a sharp peak in the low-loss region, which represents normal market conditions. However, a long heavy tail extends to the right, reflecting the risk of many defaults occurring simultaneously under severe systematic shocks.

Expected Loss Stability: The EL values are very close between Model 1 and Model 2 for both portfolios, which is expected because correlation structure mainly affects the shape and tail of the distribution rather than the mean.

Portfolio 1 Diversification Benefit: In Portfolio 1, the multi-factor model's 99% VaR is noticeably lower than the single-factor model. This difference shows the benefit of industry diversification. Although the reduction is moderate, it is still meaningful for capital allocation.

Portfolio 2 Risk Profile: Portfolio 2 also shows lower tail risk metrics under Model 2 compared to Model 1. This result surprised me because I expected the concentrated portfolio to show higher risk under the multi-factor approach. It appears that even in Portfolio 2, allowing for imperfect industry correlation helps reduce estimated extreme losses.

5. Conclusion and Portfolio Recommendations

Overall, this analysis shows that single-factor models tend to generate more conservative tail risk estimates compared to multi-factor models.

For both Portfolio 1 and Portfolio 2, I recommend adopting the multi-factor framework (Model 2) for more accurate economic capital calculations, particularly using the 99% Expected Shortfall. While industry diversification benefits are clearer in Portfolio 1, even Portfolio 2 benefits from recognizing imperfect sector correlations. Nevertheless, it is still important to monitor industry

concentration. For Portfolio 2, which has larger total exposure, setting reasonable limits on single-industry concentration would help mitigate potential losses during sector-specific downturns.

References

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